

REPORT

CUSTOMER PERSONALITY ANALYSIS USING EXPLORATORY DATA ANALYSIS

Prepared By

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1. Introduction

1.1 Project Overview

Customer Personality Analysis is a marketing dataset that contains information regarding customer demographics, purchasing behavior, spending patterns, and campaign responses. The purpose of this project is to perform Exploratory Data Analysis (EDA) to identify hidden patterns, understand customer behavior, and generate actionable business insights.

1.2 Dataset Source

Source:

<https://www.kaggle.com/datasets/imakash3011/customer-personality-analysis>

1.3 Dataset Description

The dataset contains customer demographic information such as age, education level, marital status, income, number of children, purchasing history, campaign responses, and spending across multiple product categories.

1.4 Important Variables

Variable	Description
Income	Annual household income

Education	Educational qualification
Marital_Status	Marital status
Kidhome	Number of children
Teenhome	Number of teenagers
Recency	Days since last purchase
MntWines	Amount spent on wine
MntMeatProducts	Amount spent on meat products
NumStorePurchases	Number of store purchases
NumWebPurchases	Number of web purchases
Response	Response to latest campaign

1.5 Business Questions

1. Is customer income associated with spending behavior?
 2. Which product category has the highest customer spending?
 3. How does family size affect customer spending?
 4. Which purchasing channel is preferred by customers?
 5. Which customers are more likely to accept marketing campaigns?
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2. Data Cleaning and Integrity Checks

2.1 Data Inspection

The required libraries were imported. The dataset was loaded and inspected using `info()`, `describe()`, and `head()` functions and `shape` and `columns` attributes to understand its structure and data types.

2.2 Missing Value Treatment

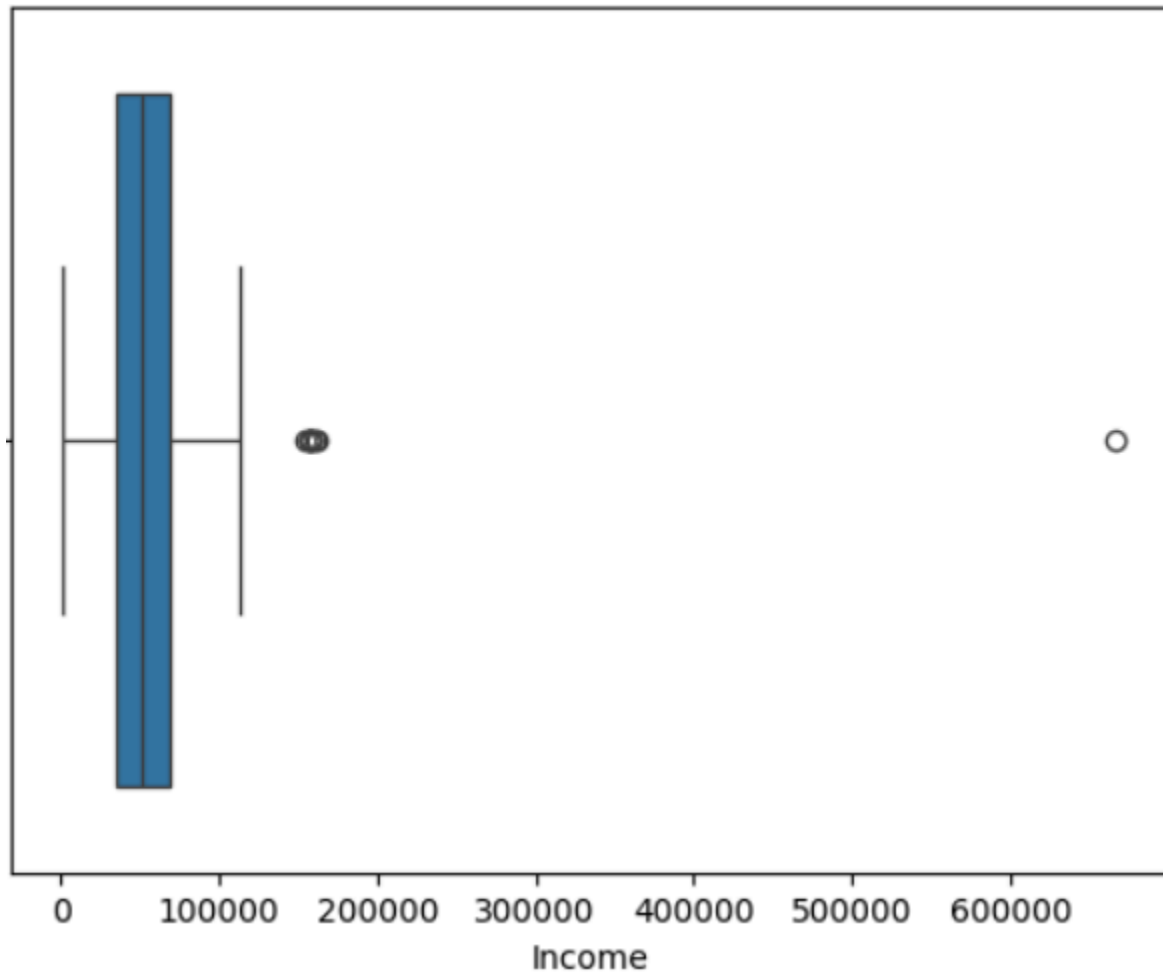
The Income column contained missing values.

I would choose median for the Income column because of the nature of the income data.

Reasons:

1. **Income usually contains outliers:** Some customers may earn very high incomes while most earn moderate incomes.
2. **Income distribution is usually skewed:** In real-world customer datasets, many people have average incomes and few people have extremely high incomes. This creates a **right-skewed distribution**. For skewed numerical data, median is preferred over mean.

Box plot representation:



So median imputation was used as income data contains outliers and skewness. The median is more robust than the mean and therefore preserves the distribution more effectively.

2.3 Duplicate Records

Duplicate records were identified and removed to ensure data quality and avoid biased analysis.

2.4 Data Type Conversion

The 'Dt_Customer' column was converted into datetime format to enable proper date-based analysis.

3. Outlier Analysis

3.1 Distribution Analysis

Numerical variables were analyzed using skewness measures and boxplots.

Several variables, particularly Income and spending-related features, exhibited positive skewness and contained extreme observations.

3.2 IQR-Based Outlier Detection

The Interquartile Range (IQR) method was applied across numerical features to identify outliers systematically.

3.3 Outlier Treatment

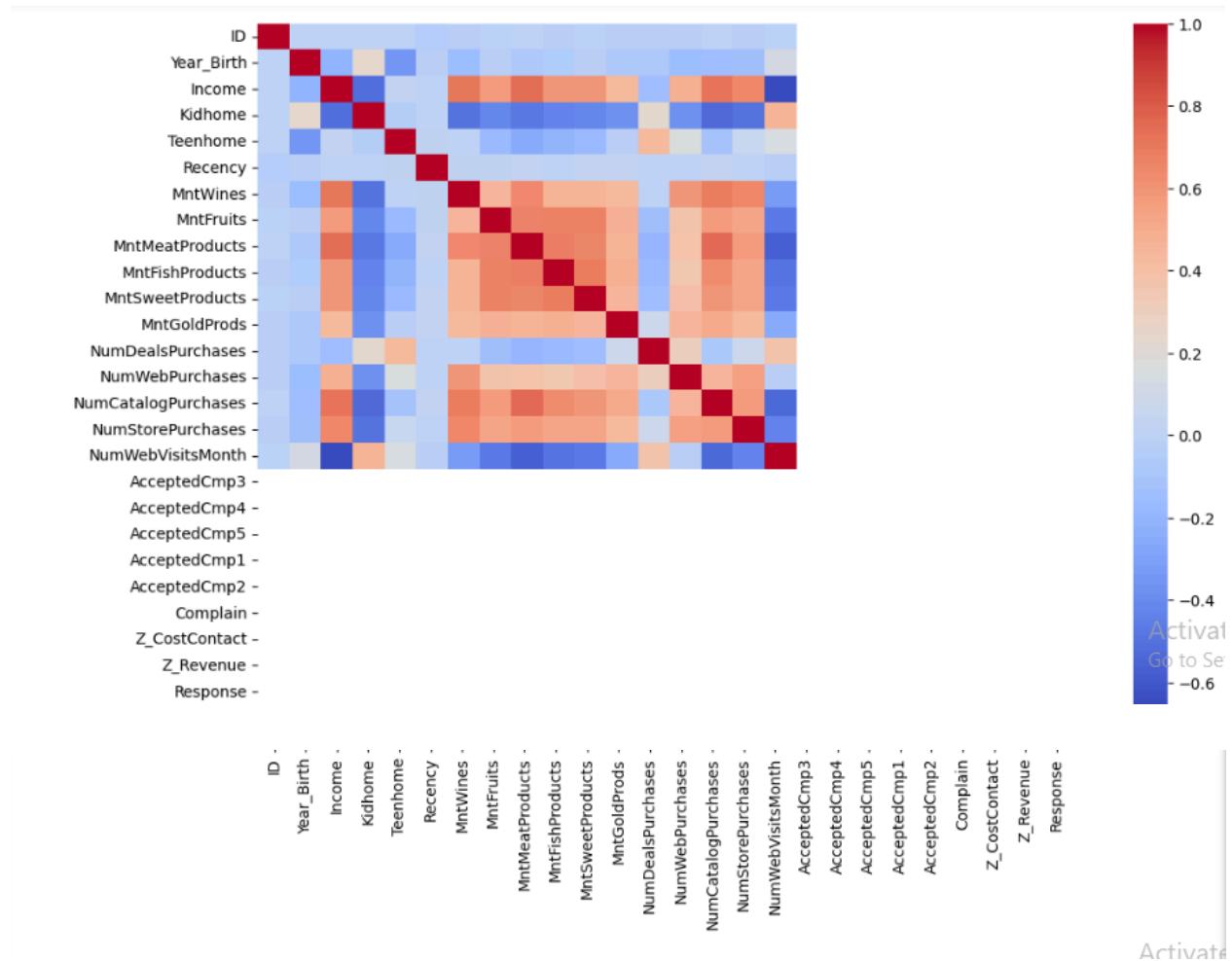
Winsorization (capping) was applied to reduce the influence of extreme values while preserving important customer information.

This approach was preferred over deleting records because high-income customers may represent valuable business segments.

4. Exploratory Data Analysis

4.1 Correlation Analysis

A correlation heatmap was constructed to study relationships among numerical variables.

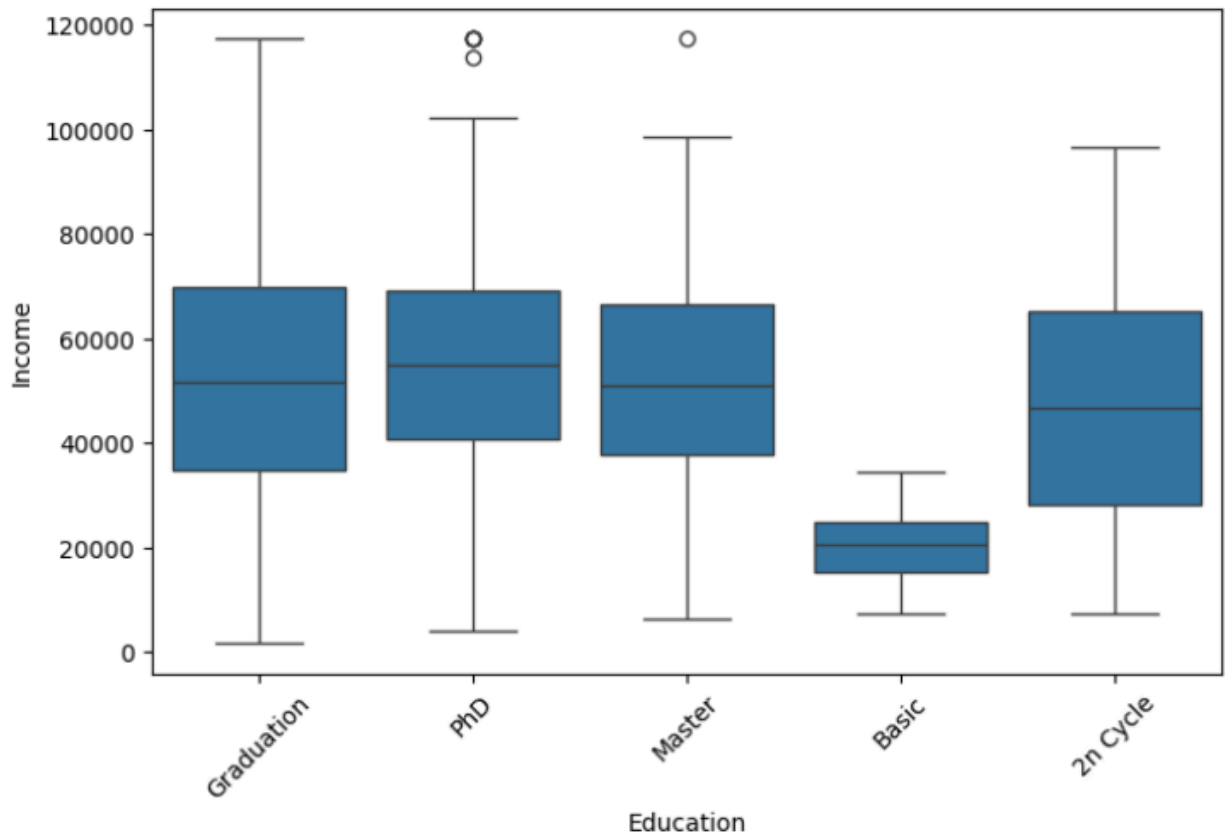


Key Findings:

- Income shows a positive relationship with customer spending.
- Customers with higher income tend to spend more across product categories.
- Catalog purchases and total spending are strongly related.
- Frequent website visits do not necessarily translate into purchases.

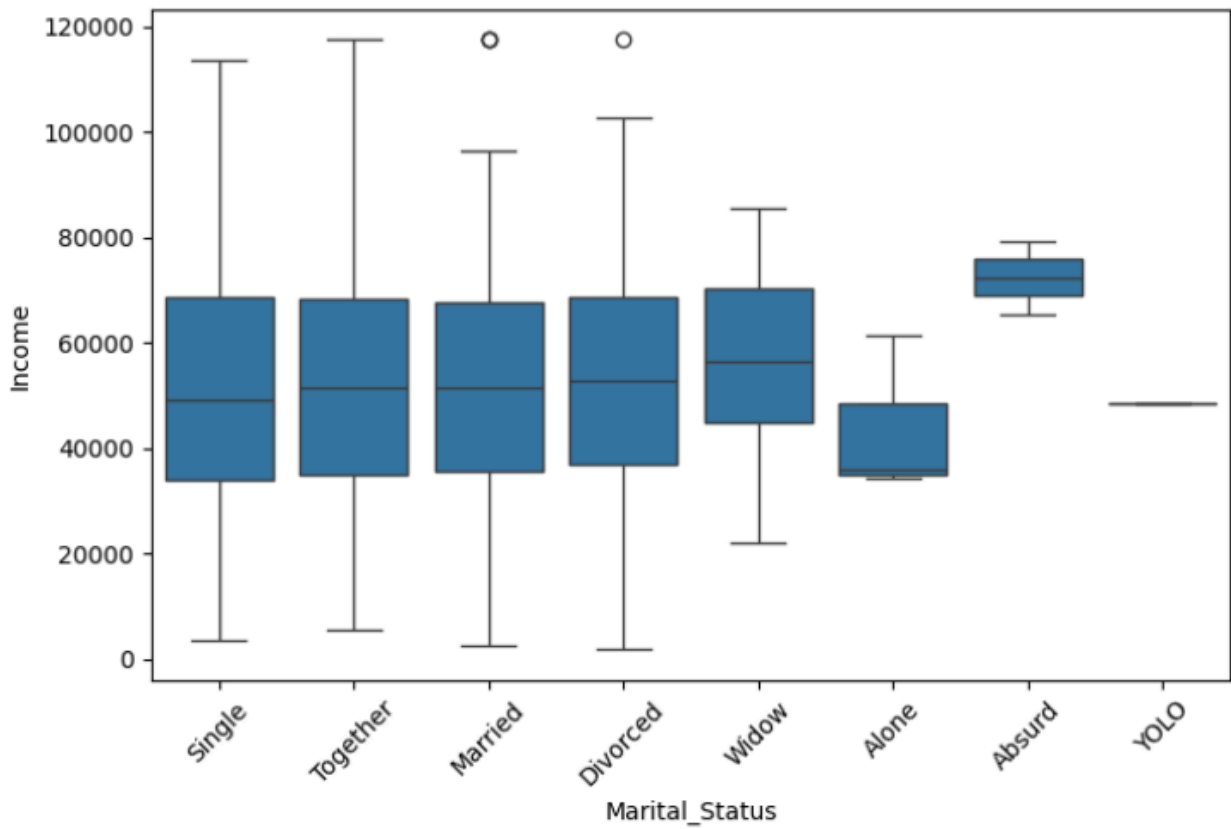
4.2 Income vs Education

Income levels vary across educational groups.



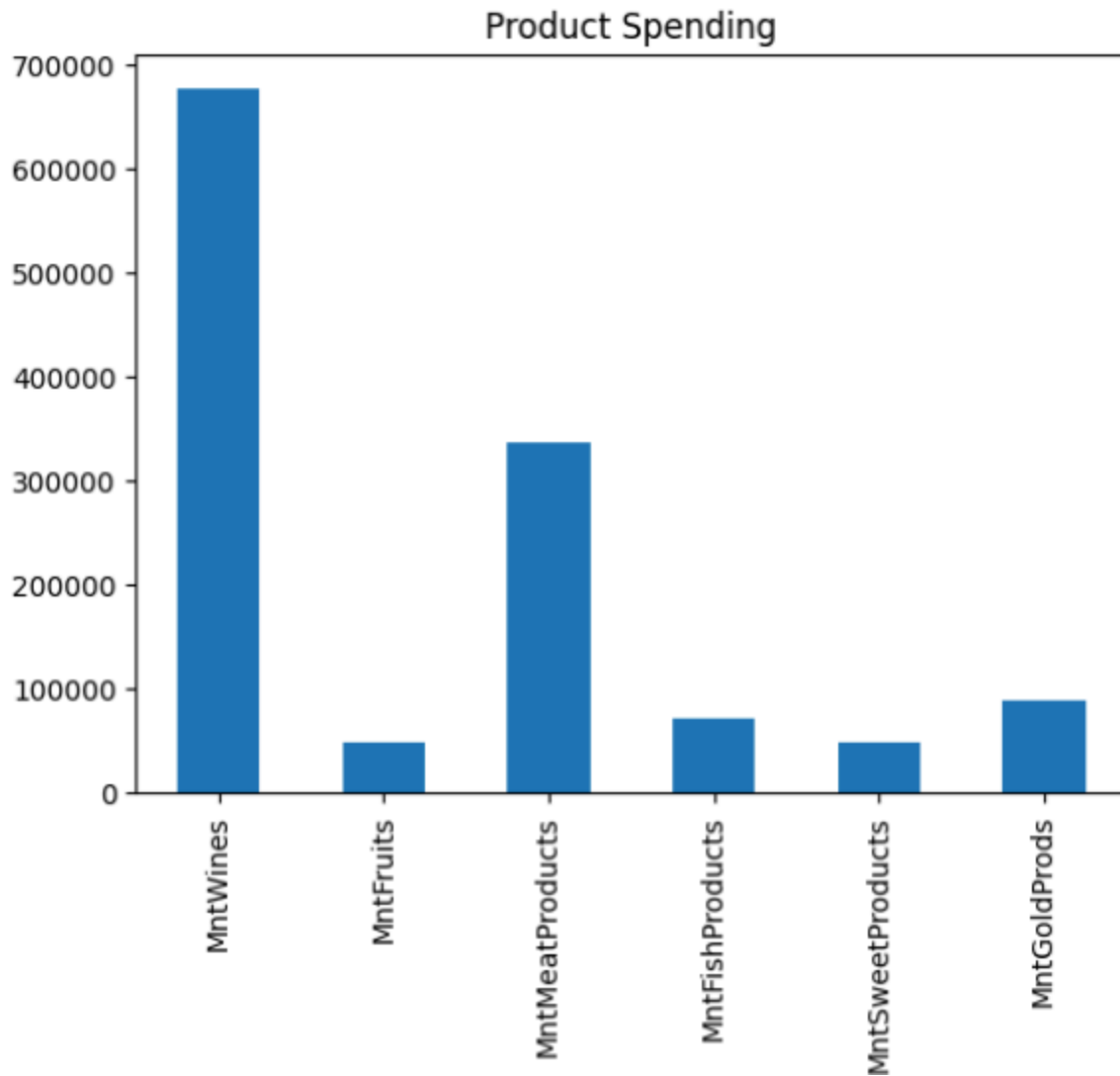
Customers with higher educational qualifications generally exhibit higher income levels.

4.3 Income vs Marital Status



Spending and income patterns differ among marital status groups, indicating the importance of demographic segmentation.

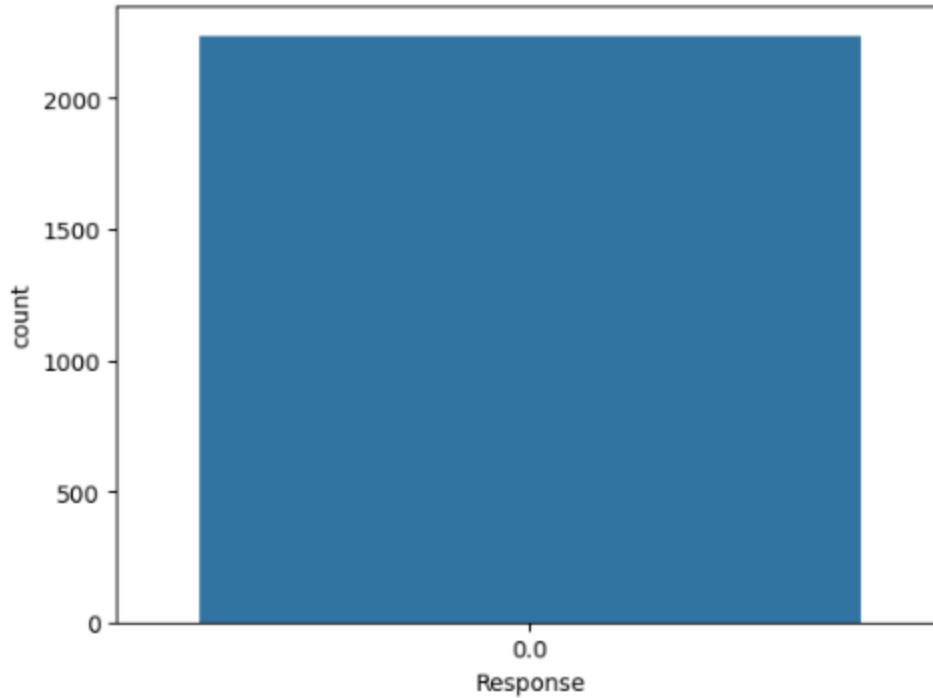
4.4 Product Category Analysis



Among all product categories, wine products contribute the highest spending amount.

This suggests that wine is a major revenue-generating category within the customer base.

4.5 Campaign Response Analysis



Most customers did not accept the latest marketing campaign.

This indicates opportunities for improving campaign targeting and personalization strategies.

Correlation vs Causation

Although several variables demonstrate strong correlations, correlation alone does not imply causation. Hidden factors such as customer lifestyle, preferences, and purchasing habits may influence observed relationships.

5. Feature Engineering and Encoding

5.1 Age Feature

A new Age feature was created using:

$\text{Age} = \text{Current Year} - \text{Year_Birth}$

This provides a more meaningful representation of customer demographics.

5.2 Total Children Feature

$\text{Total_Children} = \text{Kidhome} + \text{Teenhome}$

This feature captures overall family size.

5.3 Total Spending Feature

Total spending was calculated by combining expenditures across all product categories.

This feature enables a holistic evaluation of customer value.

5.4 Categorical Variable Identification

Ordinal Variable:

- Education

Nominal Variable:

- Marital_Status

5.5 Encoding

Education was encoded according to its natural ranking structure using **Label Encoding**.

Marital_Status was transformed using **One-Hot Encoding** to prepare the dataset for future predictive modeling.

6. Answers to Business Questions

Q1. Is customer income associated with spending behavior?

Yes. Customers with higher income generally show higher spending levels across product categories.

Q2. Which product category has the highest customer spending?

Wine products account for the highest customer spending.

Q3. How does family size affect spending?

Customers with fewer children generally exhibit higher spending patterns.

Q4. Which purchasing channel is preferred?

Store purchases are the most frequently used purchasing channel.

Q5. Which customers are more likely to accept campaigns?

Customers with higher spending levels and stronger purchasing engagement showed a higher tendency to respond positively to campaigns.

7. Conclusion

This project successfully applied Exploratory Data Analysis techniques to understand customer demographics, purchasing behavior, and marketing campaign responses.

The analysis revealed that income is a major driver of spending behavior, wine products generate the highest revenue, and purchasing preferences differ across customer segments. Feature engineering improved the analytical value of the dataset, while encoding prepared the data for future machine learning applications.

The insights obtained from this analysis can help organizations improve customer segmentation, marketing effectiveness, and business decision-making.